

LLD · INTERVIEW PREP

The Low-Level Design round.

How we evaluate the way you design scalable, reliable, and operable systems, at the class, object, and module level.

- WHAT THIS ROUND MEASURES

Not just code that runs. Code shaped to ship.

We look at how you design **scalable, reliable, and operable systems**, from the class and module level up.

Production-ready.
Not just working.

WHAT WE LOOK FOR

- 01 Clean decomposition into objects and modules.
- 02 Concurrency, handled with intent.
- 03 Resilience & recovery, built in, not bolted on.
- 04 Patterns applied where they reduce coupling.

The thinking is evaluated as much as the outcome. Walk us through it as you go.

- TOPIC SURFACE AREA

Eight areas to be fluent in.

01 OOP & SOLID fundamentals.

Encapsulation, polymorphism, inheritance, and the SOLID principles that keep code maintainable.

03 State machines & workflows.

Modelling stateful behaviour explicitly. Transitions, guards, invariants, terminal states.

05 Resilience patterns.

Timeouts, retries, circuit breakers, idempotency. Defending against failure at the call site.

07 Core design patterns.

Factory, Singleton, Strategy, Observer, Adapter, Decorator, Template Method, Repository, and when not to use them.

02 UML basics.

Class and sequence diagrams, enough to communicate a design clearly to the room.

04 Multi-threading & concurrency.

Threads, locks, race conditions, lock-free structures, the memory model of the language you use.

06 Error handling & recovery.

Classifying errors as retryable, fatal, or business; surfacing them well; recovering cleanly.

08 Testing concurrent systems.

Deterministic tests, stress tests, property-based testing. Strategies that catch races, not hide them.

- DURING THE INTERVIEW

Three habits that go a long way.

01

Think aloud.

Walk us through your reasoning as you design. We value the thought process as much as the outcome.

02

Ask clarifying questions.

Understand constraints, scale, and business requirements before diving in. There are no penalties for asking.

03

State your assumptions.

Make reasonable assumptions explicitly, so we can align or redirect early, not at the end.

MOST OF THE ROUND

This round is about **how you partner with us**, not pure technical correctness. Treat us as collaborators.

- BEFORE WE START

Pre-requisites to be ready with.



01 · CODE

An IDE or editor of your preference.

Use whatever you're fastest in. Have it running, compiling, and authenticated before the session starts. You'll be coding live on your own setup.

VS CODE

INTELLIJ

YOUR CHOICE



02 · DIAGRAM

A flowchart or diagramming tool.

Anything that lets you sketch **class diagrams and sequences quickly**. We'll be drawing alongside you.

EXCALIDRAW

WHIMSICAL

YOUR CHOICE

We'll share collaborative environment links at the start of the session.

FROM ALL OF US AT HEVO ENGINEERING

All the best.

Questions about the round, your setup, or anything else?
Reach out to your Talent partner. They're your first stop.